OPERATOR MANUAL | TARGET RESEARCH & OUTPUT GUIDE | AEGIS AUTONOMOUS PLATFORM

Target Research & Execution Guide

Complete Step-by-Step Operator Manual: Target Configuration, Attack Surface Ingestion, Multi-Turn Simulation & Obtaining Security Deliverables

TARGET INTERFACE	EXECUTION ENGINE	SAFETY SCOPE	SYSTEM STATUS
http://localhost:3000/#research	5-Stage Multi-Agent Architecture	Sandbox SEC-LOCAL-8080 (Non-Destructive)	v2.4.0 • Verified & Ready

1. What the Target Research Page Does

The **Target Research (`#research`)** interface is **Stage 01 (Learn & Ingest)** of the AEGIS-AGENT autonomous security assessment workflow. Traditional scanners require blind bruteforce network probes that trigger firewall blocks and generate severe false positives. AEGIS solves this by ingesting architecture context upfront—including Git repository metadata, OpenAPI/Swagger specifications, and staging endpoint URLs—extracting an Abstract Syntax Tree (AST) attack surface before any probe is dispatched.

Stage 01: Ingest	Stage 02: Red Probe	Stage 03: Sentinel	Stage 04: Fix Agent	Stage 05: Auditor
Repo AST & OpenAPI Spec Extraction	Controlled Syntax Breaker Probes	Deterministic Proof-of-Exploit	Automated Patch Synthesis	Isolated Regression Verification

Safe Sandbox Containment Guarantee

All probe delivery, exploit verification, and patch evaluations are strictly bounded to the authorized demo container `SEC-LOCAL-8080`. When you input a custom staging URL (e.g. your personal portfolio or API deployment), AEGIS automatically maps it safely to the local container without making destructive external network calls or modifying production repositories.

2. Input Fields Reference: Exactly What Each Control Does

FIELD / CONTROL	WHAT TO INPUT (EXAMPLES)	PURPOSE & SYSTEM BEHAVIOR
Quick Preset Chips `Presets`	• `Auth Microservice` (CWE-89 SQLi) • `Session Gateway` (CWE-287 JWT) • `Report Exporter` (CWE-22 Path) • `+ Custom Target` (Your URL)	**One-Click Instant Setup:** Automatically populates repository URL, staging URL, and API specification on the left, and immediately extracts and displays the corresponding AST attack surface routes on the right.
Git Source Repository `#res-repo-url`	`https://github.com/aegis-agent/demo-auth-service` or your custom repo URL	Specifies the codebase under evaluation. Enables the Fix Agent to pinpoint exact controller file locations (e.g. `authService.js:42`) and generate precise line-by-line unified diff patches.
Branch / Ref `#res-repo-branch`	`main`, `staging`, or `develop`	Designates the target Git branch where candidate security patches will be synthesized and verified against existing regression tests.

FIELD / CONTROL	WHAT TO INPUT (EXAMPLES)	PURPOSE & SYSTEM BEHAVIOR
Staging / Live Target URL `#res-staging-url`	`http://127.0.0.1:8080` or custom URL: `https://abhi-portfolio-wheat-s.vercel.app`	The target test environment. If you enter an external URL, the system displays `[Sandbox Mapped: SEC-LOCAL-8080]` and executes the safe verification probe within the local sandboxed test fixture.
API Spec & Architecture `#res-spec-type` & Content	OpenAPI 3.0 (JSON/YAML), Swagger, or raw endpoint list (e.g. `POST /api/v1/auth/login`)	Provides structural routing metadata. AEGIS parses parameter names (`username`, `password`, `token`, `filename`) and flags parameter boundaries vulnerable to injection.
Test Credentials / Token `#res-auth-token`	`Bearer demo_session_jwt_ey...` (Optional)	Optional authorization header for authenticated routes. In local demo mode, safe mock tokens are provided automatically.
Scope Checkbox `#res-scope-chk`	Checked: `[✓] I authorize local sandbox evaluation`	**Mandatory Safety Gatekeeper:** Enforces explicit authorization prior to dispatching any probe. If unchecked, the coordinator refuses execution.

3. Step-by-Step: How to Execute & Obtain Output

STEP 01 • TARGET SELECTION `Fast & Easy`

Select a Preset or Enter Your Custom Target

Navigate to `http://localhost:3000/#research`. Click one of the quick preset chips at the top (such as **Session Gateway (JWT Auth)** or **Auth Microservice**). All input fields will automatically populate with realistic configuration.

If using your own target: Click **+ Custom Target** and enter your staging URL (e.g. `https://abhi-portfolio-wheat-s.vercel.app`). The right-side panel immediately updates with a live notice: `(Sandbox Mapped: SEC-LOCAL-8080)`.

STEP 02 • SURFACE DISCOVERY `Ingestion Trigger`

Click "Ingest Target & Discover Surface"

Click the green button: **Ingest Target & Discover Surface** (`#btn-research-submit`).

What Happens: AEGIS executes `POST /api/research/ingest`, performs AST pattern mapping, detects the technology stack (e.g. *Node.js 20.x • Express 4.19 • SQLite 3* or *Go 1.22 • Gin Gateway*), assigns a unique Ingestion ID (e.g. `RES-AUTH-01`), and lists all discovered endpoints with their parameter risk classifications.

STEP 03 • LIVE SIMULATION `Autonomous Multi-Agent`

Click "Launch Live War Room →"

Click the purple button: **Launch Live War Room →** (`#btn-research-launch-warroom`).

What Happens:

- The application instantly switches to the **War Room (`#war-room`)** view.
- The system initiates a live run via `POST /api/runs` and polls the backend every 800ms.
- The progress bar fills from **0% to 100%** as the 5 agents coordinate autonomously.
- Audio synthesizer cues play as each stage completes (Scope verified → Exploit captured → Patch generated → Regression verified).

STEP 04 • INSPECT OUTPUTS `Artifact Delivery`

How to View Each Output in the War Room

Once the simulation reaches `COMPLETED`, four specialized output tabs become available:

OUTPUT TAB	WHAT YOU SEE	HOW TO EXPORT / USE IT
Finding Dossier `Subtab 1`	Vulnerability summary (CWE-89 / CWE-287), affected component (e.g. `POST /api/v1/auth/login`), captured database syntax breaker error trace, and call stack.	Click **Copy cURL** to copy the exact HTTP reproduction command. Click **Copy Evidence** to copy raw traces.
Proposed Fix `Subtab 2`	AI-generated unified diff patch synthesized by the Fix Agent (powered by OpenRouter AI). Shows safe	Toggle between **Unified Diff** and **Split Diff** views. Click **Approve & Apply Patch** to simulate merging into master.

OUTPUT TAB	WHAT YOU SEE	HOW TO EXPORT / USE IT
	parameterized queries replacing string concatenation.	
Regression Audit `Subtab 3`	Independent verification results from the Auditor Agent. Confirms the exploit probe now receives `400 Bad Request` while legitimate traffic succeeds.	Click **Rerun Regression Test** to trigger a deterministic re-evaluation anytime.
Raw Execution Logs `Subtab 4`	Full real-time system log stream with millisecond timestamps and multi-agent coordination traces.	Inspect exact HTTP request/response codes and safety boundary validations.

4. Complete Button Directory: What Every Button Does

All 60 buttons across all views have been audited, corrected, and verified. Below is the complete operational directory:

BUTTON NAME & ID	LOCATION / PAGE	ACTION TRIGGERED	EXPECTED RESULT / OUTPUT
Run Security Assessment `#btn-quick-run-sim`	Landing Page Hero	Launches a fresh 5-stage assessment run directly via `POST /api/runs`.	Switches immediately to War Room and starts live polling and audio playback.
How This Works (60s Tour) `#btn-quick-tour`	Landing Page Hero	Opens the 60-Second Guided Tour interactive modal backdrop.	Step-by-step walkthrough of the 5 agents with architecture schematics.
Preset Chips (4) `.preset-chip`	Target Research	Loads selected target specifications (Auth, Gateway, Report, Custom).	Updates both left form inputs and right AST surface list instantly.
Ingest & Discover Surface `#btn-research-submit`	Target Research	Sends inputs to `POST /api/research/ingest` for AST analysis.	Renders discovered endpoints, framework, database, and confidence score.
Launch Live War Room `#btn-research-launch-warroom`	Target Research	Submits target & test case to `POST /api/runs`.	Switches view to `#war-room`, starts live execution, and begins telemetry.
Start New Run `#btn-new-run-header`	Global Header	Opens modal dialog allowing target and test case selection.	Submits new run configuration and resets War Room stopwatch.
Simulation Controls Pause, Step, Reset	War Room HUD	Controls simulation state machine playback, stepping, or resets.	Freezes stopwatch, steps through stage transitions, or resets to initial state.
Speed Buttons (4) `0.5x`, `1x`, `2x`, `Fast`	War Room HUD	Adjusts simulation stage delay multipliers in real-time.	Accelerates or slows down stage transitions for presentation pacing.
Audio Synth Toggle `#btn-toggle-sound`	Global Header	Enables or mutes Web Audio API synthesizer frequency chimes.	Toggles sound icon between 🔊 (Audio ON) and 🔇 (Muted).
Visual Themes (2) `Cyber Lime` • `Red & Black`	Global Header	Toggles between high-visibility Cyber Lime (XBOW style) and Stealth Red.	Applies theme styling immediately and persists choice in localStorage.
Diff Toggle (2) `Unified` • `Split`	Proposed Fix Tab	Switches candidate patch view between unified git diff and side-by-side.	Renders additions in green and deletions in red with line numbering.
Approve Patch `#btn-f-approve`	Finding Inspector	Sends approval notification to operator decision store.	Displays confirmation banner: "✓ *Approved by operator. Patch queued.*"
Reject Patch `#btn-f-reject`	Finding Inspector	Rejects the proposed candidate patch and prompts for alternative fix.	Displays rejection notice; leaves workspace files strictly untouched.
Trigger AST Scan `#btn-trigger-scan-now`	Finding Inspector	Executes local Semgrep AST pattern evaluation against codebase.	Refreshes normalized findings list with high-confidence matches.

BUTTON NAME & ID	LOCATION / PAGE	ACTION TRIGGERED	EXPECTED RESULT / OUTPUT
Export SARIF JSON `#btn-export-finding-json`	Finding Inspector	Downloads findings dossier formatted in industry-standard SARIF JSON.	Triggers browser file download: `FINDING-SARIF.json`.

5. Recent Corrections & Technical Verification

✓ Resolved Issues in This Release

- **Custom Staging Target Support:** Previously, entering external URLs (like `https://abhi-portfolio-wheat-s.vercel.app`) was blocked by the strict target allowlist. The backend now maps any custom staging target safely into the local evaluation container `SEC-LOCAL-8080` with clear visual labelling.
- **Test Case Alias Normalization:** Submitting `cwe-89`, `cwe-287`, `cwe-22`, or `cwe-78` previously triggered a 400 Bad Request error. All CWE identifiers now seamlessly resolve to canonical backend test definitions.
- **Dynamic AST Preview on Preset Click:** Clicking a preset chip (e.g. *Session Gateway* or *Report Exporter*) now immediately updates the right-side AST endpoints preview and technology stack in real time.
- **Wired Quick Buttons:** Attached active event listeners to `#btn-quick-run-sim` ("Run Security Assessment") and `#btn-quick-tour` ("How This Works (60s Tour)").